CCNA: Introduction to Networks

Detailed Study Notes — Physical Layer & Data Link Layer

Course: CIT 3212 — Computer Networks
Topic Coverage: OSI Layers 1 & 2, Transmission Media,
Signal Impairments, Error Detection & Correction

Expanded from the lecture slide deck “CCNA: Introduction to Networks”

These notes explain, extend and organise the original slides into a study-ready reference.

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1. Networking Fundamentals & the OSI Model

A **computer network** is a set of two or more devices connected so that they can exchange data and share resources (files, printers, internet access). For two devices to communicate reliably, they must agree on a common set of rules called **protocols**. To manage this complexity, the communication process is broken into layers — this is the idea behind the **OSI (Open Systems Interconnection) reference model**.

OSI Model (7 layers, bottom → top):

1. **Physical** — raw bits over a medium *(covered in these notes)*
2. **Data Link** — framing, MAC addressing, error detection *(covered in these notes)*
3. Network — logical addressing & routing (IP)
4. Transport — end-to-end delivery, reliability (TCP/UDP)
5. Session — dialog control
6. Presentation — data format, encryption, compression
7. Application — user services (HTTP, FTP, SMTP)

Each layer provides services to the layer above it and consumes services from the layer below it. The **encapsulation** process adds a header (and sometimes a trailer) at each layer as data moves down the stack on the sender, and each header is removed as data moves up on the receiver.

Why layers matter

- **Modularity** — a change in one layer (e.g. swapping copper for fibre) does not require redesigning the others.
- **Interoperability** — vendors implement to the same layer standards, so equipment from different manufacturers works together.
- **Troubleshooting** — problems can be isolated to a single layer.

These notes focus on the two lowest layers: the **Physical Layer (Layer 1)** and the **Data Link Layer (Layer 2)**.

2. The Physical Layer (Layer 1)

2.1 Definition

The **Physical Layer** is the lowest layer of the OSI model. It is responsible for transmitting **raw data bits** (0s and 1s) through the physical network hardware, converting them into signals that can travel across a physical medium.

At this layer there is no concept of addresses, packets or frames — only individual bits and the electrical, optical or radio signals that represent them.

2.2 Core responsibilities

- Handles **physical and electrical** data transmission.
- Includes **cables, connectors, plugs, pins, receivers and transceivers**.
- **Transfers raw bits** between devices over the medium.
- Defines **hardware standards** — voltages, pin layouts, cable types, signal types, connector shapes.
- Defines the **data rate** (bits per second) and how the transmission is synchronised.

2.3 Functions of the Physical Layer

(a) Bit-by-bit transmission

The layer sends data as a **stream of individual bits**, not as blocks. Each bit is represented by a physical state — for example a voltage level on copper, a pulse of light in fibre, or a change in a radio wave.

(b) Encoding and decoding

Encoding is the process of converting the data (bits) into a signal suitable for the medium. **Decoding** is the reverse — recovering the original bits from the received signal at the destination.

> Example line-coding schemes include **NRZ (Non-Return-to-Zero)**, **Manchester** and **4B/5B**. Manchester encoding embeds the clock into the signal, helping the receiver stay synchronised.

(c) Signal transmission

The encoded signal is carried across the medium. The nature of the signal depends on the medium: electrical voltage/current on copper, light pulses in fibre optics, and electromagnetic waves in wireless.

(d) Modulation and demodulation

- **Modulation** — modifying a carrier signal (amplitude, frequency or phase) so that it can carry the data over a medium, especially for long distances or wireless transmission.
- **Demodulation** — extracting the original data from the received modulated signal.
- A **modem** (MOdulator–DEModulator) performs both operations.

(e) Transmission modes

The Physical Layer decides the direction of data flow:

Mode	Direction	Description	Example
Simplex	One-way only	Transmission is always in one direction; one device always sends, the other always receives.	Keyboard to computer; TV broadcast
Half-duplex	Two-way, alternately	Both devices can send and receive, but not at the same time.	Walkie-talkie; older Ethernet hubs
Full-duplex	Two-way, simultaneously	Both devices can send and receive at the same time.	Telephone call; switched Ethernet

(f) Data-rate control

The layer controls the **speed** (bits per second) and **timing/synchronisation** of transmission so that sender and receiver remain in step. Mismatched rates or timing cause loss of synchronisation and errors.

> **Key idea:** Layer 1 is concerned only with *how to move a bit* — the physics, the signal and the timing. Everything about *meaning*, *addressing* and *reliability* happens in higher layers.

3. Transmission Media

> **Transmission media** is the physical or wireless communication channel used to carry data signals from one device to another in a network. It is the fundamental pathway that provides connectivity between networked devices.

3.1 Factors affecting the choice of a medium

- **Transmission distance** — how far the signal must travel.
- **Data transfer speed (bandwidth)** — how much data per second is required.
- **Susceptibility to interference and noise** — the environment (electrical machinery, weather, obstacles).
- **Cost and installation requirements** — budget, labour, and difficulty of installation.

3.2 Classification of transmission media

Category	Also called	Signal path	Examples
Guided (bounded)	Wired media	Signal is confined to a physical cable along a fixed route	Twisted pair, coaxial, optical fibre
Unguided (unbounded)	Wireless media	Signal propagates freely through air, vacuum or water	Radio, microwave, infrared

3.3 Guided media

Guided media (wired/bounded) transmit data signals through a physical path using cables. The signal is **confined and guided** along a fixed route, giving controlled communication between devices.

- Uses physical cables to transmit signals.
- Provides dedicated, well-defined transmission paths.
- Offers **higher data rates** than most wireless media.
- Provides **better security** — physical access is required and signal leakage is limited.
- Suitable for short to long distances depending on cable type.

Three major types: **Twisted Pair**, **Coaxial** and **Optical Fibre** cables.

3.3.1 Twisted Pair Cable

A twisted pair cable consists of **two individually insulated copper conductors twisted together in a helical pattern**. The twisting cancels out much of the electromagnetic

interference (EMI) and crosstalk between adjacent pairs, because the two wires pick up nearly identical noise which then cancels at the receiver. Several twisted pairs are usually bundled inside a protective outer sheath.

- Twisting of conductors reduces interference and signal degradation.
- Multiple pairs can be bundled within one cable.
- Low cost, flexible and easy to install — the most widely used guided medium.
- Common in telephone systems and LANs.

Unshielded Twisted Pair (UTP)

UTP consists of twisted copper wire pairs **without any additional metallic shielding**. The twisting alone provides basic protection against interference, making UTP suitable for environments with moderate noise levels.

Advantages

- ✔ Least expensive guided medium.
- ✔ Easy to install and maintain (lightweight, flexible).
- ✔ Supports high data rates over short distances.
- ✔ Widely used in Ethernet LANs and telephone networks.

Disadvantages

- ✘ Lower performance/capacity than STP.
- ✘ Limited distance due to attenuation.
- ✘ More susceptible to noise and EMI.

UTP is commonly categorised as Cat 3, Cat 5e, Cat 6, Cat 6a etc.; higher categories support higher bandwidth. Standard maximum run for Ethernet UTP is 100 m.

Shielded Twisted Pair (STP)

STP consists of twisted pairs of insulated copper conductors enclosed within an **additional metallic shielding layer** — foil, braid, or both. This shielding gives enhanced protection against external EMI and reduces crosstalk, resulting in more stable and reliable transmission.

Advantages

- ✔ Better performance at higher data rates than UTP.
- ✔ Significantly reduces EMI and crosstalk.
- ✔ More reliable in electrically noisy environments.
- ✔ Suited to high-speed Ethernet and voice/data systems.

Disadvantages

- ✘ Higher cost than UTP.
- ✘ More complex installation and termination (shielding must be grounded).
- ✘ Bulkier and less flexible — harder cable management.

3.3.2 Coaxial Cable

Coaxial cable consists of a **central copper conductor** surrounded by a **dielectric insulating layer**, then a **metallic shielding layer**, and finally an **outer protective jacket**. This layered structure gives much better protection against noise and interference than twisted pair.

- Supports **baseband** (single channel) and **broadband** (multiple channels) operation.
- Provides **higher bandwidth** than twisted pair.
- Strong shielding makes it resistant to noise, crosstalk and EMI.
- Commonly used in cable TV (CATV), broadband networks and analogue TV.
- Durable and reliable due to its layered construction; easier to install than fibre.

Advantages

- ✔ Higher bandwidth than twisted pair.
- ✔ Better signal quality and reliability.
- ✔ Allows multiple channels using frequency-division multiplexing (FDM).
- ✔ Less affected by external EMI.

Disadvantages

- ✘ More expensive than twisted pair.
- ✘ Requires proper grounding for safety and to minimise interference.
- ✘ Bulkier and less flexible due to multiple layers.
- ✘ Can be physically tapped — a security risk.

3.3.3 Optical Fibre Cable

Optical fibre transmits data as **light signals** through a glass or plastic **core**, using the principle of **total internal reflection**. The core is surrounded by a **cladding** layer of lower refractive index that confines the light within the core, enabling very high-speed, long-distance transmission.

- Supports very large data volumes at extremely high speeds.
- Can operate in unidirectional or bidirectional modes.
- **WDM (Wavelength Division Multiplexing)** lets multiple light signals (different colours/wavelengths) travel simultaneously over a single fibre.
- Used wherever high bandwidth, long distance and low signal loss are required.
- Two main types: **single-mode** (one light path, very long distance) and **multimode** (multiple light paths, shorter distance).

Advantages

- ✔ Very high bandwidth and data-carrying capacity.
- ✔ Lightweight and compact — easier long-distance installation.
- ✔ Low signal attenuation over long distances.
- ✔ Immune to EMI and electrical noise.
- ✔ Resistant to corrosion and harsh environments.

Disadvantages

- ✘ Complex installation/maintenance (precise splicing and handling).
- ✘ High initial installation and equipment cost.
- ✘ Fragile compared to copper — needs careful handling.

Applications of optical fibre

Field	Use
Medical	Endoscopy, imaging systems, laser-based surgery
Defence & Aerospace	Secure, high-speed, interference-free communication
Communication	Internet backbones, undersea cables, telecom systems
Industrial & Automotive	Sensing, lighting, data communication, safety systems

3.4 Unguided (Wireless) Media

Unguided media — also called **wireless** or **unbounded** media — uses electromagnetic waves to transmit data **without any physical medium**. Signals propagate through free space such as air or vacuum. The three main types are **radio waves, microwaves and infrared waves**.

General features:

- Signals propagate through air or free space.
- Less secure due to their broadcast nature.
- Suitable for long-distance communication.

3.4.1 Radio Waves

Radio waves are electromagnetic waves that are **easy to generate** and can **propagate through buildings and obstacles**. They do **not require line-of-sight** between antennas, making them ideal for broadcast communication and wireless data transmission.

- **Frequency range:** 3 kHz – 300 GHz.
- **Applications:** AM/FM radio, television, cordless phones, wireless communication.
- **Types:** Shortwave (AM radio broadcasting), VHF (FM radio and TV), UHF (TV and mobile communication).
- **Components:** a *transmitter* that generates and modulates the signal and a *receiver* that receives and demodulates it.

3.4.2 Microwaves

Microwaves are an unguided medium that uses **line-of-sight** communication: the transmitting and receiving antennas must be properly aligned. The transmission range depends on the height of the antennas.

- **Frequency range:** 1 GHz – 300 GHz.
- **Applications:** mobile communication, satellite links, television distribution.

Advantages

- ✔ Cost-effective compared to laying cables.
- ✔ No need for land acquisition.
- ✔ Suitable over difficult terrain and oceans.
- ✔ Supports high data rates.

Disadvantages

- ✘ Less secure without encryption.
- ✘ Susceptible to weather (rain, fog).
- ✘ Affected by obstacles (line-of-sight requirement).
- ✘ High cost of antenna design, installation and maintenance.

3.4.3 Infrared

Infrared waves are used for **short-range** wireless communication. They **cannot penetrate solid obstacles**, which limits their range but helps minimise interference between nearby systems.

- **Frequency range:** 300 GHz – 400 THz.
- **Applications:** TV remote controls, wireless keyboards, mice, printers.

3.5 Comparison: Radio vs Microwave vs Infrared

Feature	Radio Waves	Microwaves	Infrared Waves
Directionality	Omnidirectional	Highly directional	Directional; requires line of sight (LOS)
Obstacle penetration	Penetrates buildings/obstacles easily (low frequency)	Poor penetration; LOS required	Cannot penetrate obstacles
Frequency range	3 kHz – 300 GHz	1 GHz – 300 GHz	300 GHz – 400 THz
Security	Low	Medium (encryption possible)	Relatively high, but limited range
Attenuation	Moderate	Variable (weather-dependent)	High
Licensing	Some bands require government licence	Most bands licensed	Generally unlicensed
Cost	Low to moderate	High	Low
Distance	Long-distance	Long-distance	Very short distance only

4. Transmission Impairment

> **Transmission impairment** is the loss or distortion of signals during transmission, leading to errors or reduced quality in communication. The three main causes are **attenuation**, **distortion** and **noise**.

4.1 Attenuation

Loss of signal strength as the signal propagates over the medium, caused by resistance and energy loss. Over distance, the signal becomes weaker and may become unreadable.

- **Analogue systems:** *amplifiers* are used to boost signal strength (but they also amplify noise).
- **Digital systems:** *repeaters / regenerators* are used to restore the original signal.

4.2 Distortion

Occurs when the **shape** of the transmitted signal changes. It typically affects **composite signals** in which different frequency components travel at different speeds, causing **phase shifts** and timing differences at the receiver. The signal's form is altered even though its strength may be fine.

4.3 Noise

Unwanted electrical or electromagnetic signals that interfere with the original signal, potentially corrupting data. Common types:

Type of noise	Cause
Thermal noise	Random motion of electrons in conductors (also called white/Johnson noise).
Induced noise	External sources such as motors and appliances inducing currents in the medium.
Crosstalk	One wire's signal coupling into a neighbouring wire.
Impulse noise	Sudden spikes from lightning, power surges or switching — a major cause of burst errors.

4.4 Factors considered when designing transmission media

- **Bandwidth** — the range of frequencies a medium can support. Higher bandwidth allows higher data rates, provided noise and attenuation remain constant.
- **Transmission impairment** — occurs when the received signal differs from the transmitted signal; signal quality suffers as a result.
- **Interference** — disturbance caused by unwanted external signals (electromagnetic radiation, neighbouring channels). It degrades signal clarity and may corrupt data.

4.5 Applications of transmission media

Transmission medium	Typical application
Unshielded Twisted Pair (UTP)	LANs, telephones
Shielded Twisted Pair (STP)	Industrial networks, high-interference environments
Optical Fibre Cable	Long-distance communication, internet backbones
Coaxial Cable	Cable TV, broadband internet, CCTV
Radio	Wireless communication, AM/FM radio, mobile phones
Infrared	Remote controls, short-range communication
Microwave	Satellite communication, radar, long-distance links

5. The Data Link Layer (Layer 2)

5.1 Definition

> The **Data Link Layer** is the second layer from the bottom in the OSI model. It organises data into **frames** and provides **node-to-node delivery** over a physical link. It also provides an interface between the Network Layer (above) and the Physical Layer (below).

5.2 Major functions

- **Framing** — dividing the bit stream into manageable, addressable units called frames.
- **Physical (MAC) addressing** — identifying the sender and receiver on the local link.
- **Error detection (and correction)** — detecting bits corrupted in transit.
- **Flow control** — preventing a fast sender from overwhelming a slow receiver.
- **Media access control** — deciding which device may transmit on a shared medium.

5.3 Sub-layers of the Data Link Layer

The Data Link Layer is divided into two sub-layers:

1. Logical Link Control (LLC)

The upper sub-layer. It provides services to the Network Layer and handles:

- **Protocol identification** — identifies which network-layer protocol (e.g. IP) is carried and ensures correct delivery.
- **Flow control** — regulates the rate of transmission to prevent data overflow between sender and receiver.
- **Error control** — detects transmission errors and may request retransmission of corrupted frames.
- Also handles multiplexing and acknowledgements depending on the LLC service used.

2. Media Access Control (MAC)

The lower sub-layer. It controls how devices access the shared physical medium:

- **Physical addressing** — adds the MAC address to frames to uniquely identify sender and receiver devices.
- **Media access management** — controls when a device may send, to avoid collisions on shared channels.
- **Frame transmission** — encapsulates packets from the Network Layer into frames and passes them to the Physical Layer.

> **Remember:** *LLC* = logical, protocol and reliability functions; *MAC* = physical addressing and who-gets-to-talk functions.

6. Error Detection & Correction

6.1 Types of errors

Whenever bits flow from one point to another, interference can change the shape of the signal and corrupt the data. Two main error types exist:

Error type	Definition	Notes
Single-bit error	Only **one bit** of a data unit (byte, character, packet) is changed from 1 to 0 or 0 to 1.	Rare in serial transmission; easier to detect.
Burst error	**Two or more bits** in the data unit are changed.	Common with impulse noise; the affected bits need not be consecutive.

6.2 Redundancy — the central concept

> To detect or correct errors we send **extra (redundant) bits** along with the data. These bits are added by the sender and removed by the receiver. Their presence lets the receiver detect — or even correct — corrupted bits.

Redundancy increases overhead but is the foundation of all error-detection schemes (parity, checksum, CRC, etc.).

6.3 Error detection techniques

(a) VRC — Parity Bit (Vertical Redundancy Check)

- A k-bit dataword is changed to an n-bit codeword where $n = k + 1$.
- The extra bit, called the **parity bit**, is chosen so that the total number of 1s in the codeword is **even** (even parity).
- **Even parity:** add a bit so the count of 1s is even. **Odd parity:** make the count odd.

> Example — data `1011001` has four 1s (already even). Even parity → append `0` → `10110010`.
> Data `1010001` has three 1s (odd). Even parity → append `1` → `10100011`.

Limitation: VRC detects only an odd number of bit errors. If two bits flip, the parity still matches and the error goes undetected.

(b) Checksum

Used mainly in the Transport Layer (e.g. IP/TCP/UDP). A block of data is treated as a list of numbers; the sender computes their **sum** and appends it to the data.

> If the data numbers are `(7, 11, 12, 0, 6)`, the sender transmits `(7, 11, 12, 0, 6, 36)`, where 36 is the sum. The receiver re-adds the five numbers and compares the total with the received sum. If they match, the data is accepted; otherwise an error is assumed and the data is discarded.

(c) LRC — Longitudinal Redundancy Check

- A block of bits is organised into a table (rows and columns).
- A **parity bit is calculated for each column**, producing a new row of parity bits for the whole block.
- The first parity bit is calculated from all the first bits of each row, and so on.
- The row of parity bits is attached to the data and sent; the receiver recomputes and compares.

LRC is stronger than simple VRC because it can detect many burst errors, but it still fails for some patterns.

(d) CRC — Cyclic Redundancy Check

- The most powerful of the redundancy-checking techniques; used in Ethernet, WLAN and many link protocols.
- Unlike VRC and LRC, which are based on addition, CRC is based on **binary division**.
- A sequence of redundant bits — the **CRC remainder** — is appended to the data so that the resulting data unit is **exactly divisible by a predetermined binary number (the generator polynomial)**.
- At the receiver, the data unit is divided again. If the remainder is zero, no error is assumed; a non-zero remainder means the data was **damaged in transit and must be rejected**.

6.4 Error correction

Two broad approaches:

Approach	Mechanism	Notes
ARQ (Automatic Repeat reQuest) — sender-based	Detect the error, then ask the sender to retransmit the data.	Requires a feedback channel; the common practical method.
FEC (Forward Error Correction) — receiver-based	The receiver works out what the correct data probably was, using error-correcting codes such as **Hamming code**.	No retransmission needed, but requires lots of redundancy (overhead).

> In theory any detected error can be corrected automatically, but this is not done in practice because it requires a great deal of redundancy (overhead). Practical systems usually *detect* with CRC/checksum and *correct* by retransmission (ARQ).

7. Framing, Flow Control, Addressing & Layer-2 Devices

7.1 Framing

The Data Link Layer wraps the Network-Layer packet in a frame: it adds a **header** (containing the destination and source MAC addresses) and a **trailer** (containing error-detection information such as a CRC/FCS). Framing lets the receiver determine exactly where one unit of data begins and ends, and identifies the devices involved in the local link.

> Typical frame structure: `[ Preamble | Dest MAC | Src MAC | Type/Length | Data (payload) | FCS ]`

7.2 Flow control

Flow control keeps a fast transmitter from flooding a slower receiver. If the receiver's buffer fills up, frames are lost. Techniques include **stop-and-wait** and **sliding-window** flow control, where the receiver signals the sender how much more data it can accept.

7.3 Addressing

At Layer 2, devices are identified by a **MAC address** — a 48-bit (6-byte) hardware address burned into the network interface card (NIC), usually written in hexadecimal (e.g. `00:1A:2B:3C:4D:5E`). The first 3 bytes are the **OUI** (Organisationally Unique Identifier, the vendor) and the last 3 bytes identify the specific device. MAC addresses are used only within the local network segment; routers change the Layer-2 addressing as packets cross networks.

7.4 Devices operating at the Data Link Layer

Device	Function at Layer 2
NIC (Network Interface Card)	Provides the MAC address and frames/deframes data.
Switch	Forwards frames based on MAC addresses using a MAC address table; each port is a separate collision domain.
Bridge	Connects two LAN segments and filters/forwards frames by MAC address.
Access Point (Wireless)	Uses MAC-layer (802.11) framing and media access for wireless clients.

8. Summary, Glossary & Revision Questions

8.1 Quick summary

Layer	Unit	Key job	Key terms
Physical (1)	Bit	Move raw bits as signals over a medium	Encoding, modulation, transmission modes, media
Data Link (2)	Frame	Node-to-node delivery, framing, error detection, MAC addressing	LLC, MAC, CRC, parity, flow control

8.2 Glossary

Term	Meaning
Bandwidth	Range of frequencies a medium can carry; upper limit on data rate.
Attenuation	Loss of signal strength over distance.
Distortion	Change in the shape of a composite signal due to differing propagation speeds.
Noise	Unwanted signals that corrupt data.
Crosstalk	Signal from one pair/wire leaking into another.
Modulation	Varying a carrier to carry data.
Multiplexing	Combining several signals onto one medium (FDM, WDM, TDM).
Redundancy	Extra bits added to allow error detection/correction.
Frame	Layer-2 data unit with header and trailer.
MAC address	48-bit hardware address of a NIC.

8.3 Revision questions

1. List and explain the main functions of the Physical Layer.
2. Distinguish between guided and unguided transmission media, giving two examples of each.
3. Compare UTP and STP cables in terms of cost, performance and interference.
4. Explain total internal reflection and how it enables optical fibre transmission.
5. Differentiate radio waves, microwaves and infrared in terms of directionality, penetration and range.
6. Define attenuation, distortion and noise. How is each countered?
7. Describe the two sub-layers of the Data Link Layer and their functions.
8. Differentiate between single-bit and burst errors.
9. Explain how a parity bit (VRC) detects errors, and state its main limitation.
10. Why is CRC considered more powerful than VRC and LRC?
11. Compare ARQ and FEC as error-correction strategies.
12. What is a MAC address, and how is it structured?

End of notes. Compiled from the lecture deck “CCNA: Introduction to Networks” (CIT 3212 — Computer Networks), with explanatory expansions added for study use.